

ALADIN AI PRODUCT ROADMAP

Voice AI Commerce + Vietnamese OCR + AI Agent Automation

PRODUCT VISION

"Order in 10 seconds, not 10 minutes"

Transform rural shop owners from spending 2 hours/day on phone orders to:

-  **Voice ordering** in Vietnamese dialect (10 seconds)
 -  **Photo ordering** - snap supplier invoice, auto-reorder (5 seconds)
 -  **AI agent** handles everything: recommendations, fulfillment, notifications
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CORE FEATURES

1. Voice AI Commerce (Zalo Integration)

Shop owner speaks in Vietnamese → AI understands → Places order → Confirms via voice

2. Vietnamese OCR (Photo-to-Order)

Shop owner photos their old invoice → AI extracts products → Suggests reorder → 1-tap confirm

3. AI Agent (Autonomous Operations)

- Predicts when shop will run out of stock → Auto-suggests reorder
 - Learns preferences → Personalized recommendations
 - Handles entire order lifecycle → Zero human intervention
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12-MONTH ROADMAP

PHASE 1: FOUNDATION (Months 1-3)

MVP - Prove the Concept

Month 1: Basic Zalo Chatbot (No AI Yet)

What: Simple keyword-based ordering via Zalo **Why:** Get users comfortable with chat ordering before adding AI

Features:

-  Menu-based ordering (numbered options)
-  Order history lookup
-  Order status tracking
-  Basic product search (keyword matching)

Tech Stack:

- Zalo Mini App / Zalo OA (Official Account)
- Backend: Node.js + PostgreSQL
- No ML/AI yet (if-then logic)

Success Metric: 40% of orders via Zalo (vs phone)

Month 2: Voice Input (Vietnamese Speech-to-Text)

What: Shop owners can speak orders instead of typing **Why:** Many shop owners struggle with typing, voice is faster

Features:

-  Voice message → Text conversion (Zalo built-in)
-  Vietnamese language support (Southern dialect priority)
-  Order confirmation via voice message
-  Fallback to menu if speech unclear

Tech Stack:

- Zalo Voice API (built-in speech recognition)
- Google Cloud Speech-to-Text (backup)
- Intent matching: Basic NLP (spaCy Vietnamese)

Success Metric: 20% of orders via voice (50 customers testing)

Month 3: Vietnamese OCR (Invoice Photo → Order)

What: Shop owners photo old supplier invoices → AI extracts products **Why:** Easiest ordering method - "Order same as last time"

Features:

-  Photo upload via Zalo
-  OCR extracts: Product names, quantities, prices
-  Match products to Aladin catalog (fuzzy matching)
-  1-tap confirmation to reorder

Tech Stack:

- Google Cloud Vision API (Vietnamese OCR)
- PaddleOCR (Vietnamese optimized, self-hosted)
- Product matching: String similarity (FuzzyWuzzy)
- Fallback: Manual correction UI

Success Metric: 80% OCR accuracy, 15% of orders via photo

Phase 1 Goal: 60% of orders via Zalo (voice + photo + chat)

PHASE 2: AI INTELLIGENCE (Months 4-6)

Smart Agent - Predict & Recommend

Month 4: Demand Forecasting AI

What: Predict when shop will run out of stock **Why:** Proactive ordering = never run out = customer loyalty

Features:

-  Analyze order history (frequency, quantity, seasonality)
-  Predict stockout date: "Your rice will run out in 3 days"
-  Auto-suggest reorder 2 days before stockout
-  1-tap to confirm suggested order

AI Model:

- Time-series forecasting (Prophet or LSTM)
- Input: 3+ months order history per customer
- Output: "Order [Product X] by [Date]"

Success Metric: 70% prediction accuracy, 30% customers adopt auto-suggestions

Month 5: Personalized Recommendations

What: AI suggests products shop owners didn't know they needed **Why:** Increase AOV, cross-sell, upsell

Features:

-  "Customers like you also bought..."
-  Seasonal recommendations (e.g., mooncakes before Mid-Autumn)
-  Bundle deals: "Buy rice + oil, save 5%"
-  New product alerts based on shop profile

AI Model:

- Collaborative filtering (similar shops)
- Content-based filtering (product categories)
- Hybrid recommendation engine

Success Metric: 20% of customers try recommended products, AOV +15%

Month 6: Voice AI Agent (Conversational Ordering)

What: Natural conversation in Vietnamese - AI understands context **Why:** "Order like talking to a human"

Features:

- ☒ Multi-turn conversation:
 - User: "Tôi cần gạo" (I need rice)
 - AI: "Loại gạo nào? ST25 hay Jasmine?" (Which rice?)
 - User: "ST25, 10 bao" (ST25, 10 bags)
 - AI: "Xác nhận 10 bao ST25, 2.5 triệu đồng?" (Confirm?)
- ☒ Context awareness (remembers previous orders)
- ☒ Handles corrections: "Không, 15 bao" (No, 15 bags)
- ☒ Vietnamese slang/dialect understanding (Saigon, Mekong)

AI Model:

- LLM: Claude API or GPT-4 (Vietnamese fine-tuned)
- Intent classification + Entity extraction
- Dialogue state tracking

Success Metric: 85% conversation success rate, 50% of orders via voice AI

Phase 2 Goal: AI handles 40% of ordering decisions autonomously

PHASE 3: FULL AUTOMATION (Months 7-9)

Autonomous Agent - Zero Human Effort

Month 7: Auto-Reorder System (Set-It-Forget-It)

What: AI automatically places orders based on predicted stockouts **Why:** Shop owners do NOTHING, orders happen automatically

Features:

- ☒ Shop owner opts in: "Auto-order rice every 2 weeks"
- ☒ AI monitors stock levels (based on sales + history)
- ☒ Auto-generates order 2 days before stockout
- ☒ Sends confirmation: "We'll deliver rice tomorrow, ok?"
- ☒ Shop owner can adjust/cancel anytime

-  Safety: Spending limit (e.g., max 5M VND/week)

AI Model:

- Reinforcement learning (optimize reorder timing)
- Inventory management algorithm
- Anomaly detection (catch unusual patterns)

Success Metric: 40% of shops on auto-reorder, 95% satisfaction

Month 8: Smart Fulfillment Optimization

What: AI optimizes entire supply chain (warehouse → delivery) **Why:** Lower costs, faster delivery, better margins

Features:

-  **Warehouse:** AI predicts demand → Pre-stocks inventory
-  **Routing:** AI optimizes delivery routes (20 stops → 30 stops/day)
-  **Batching:** Groups orders from same zone for efficiency
-  **Dynamic pricing:** Surge pricing for rush orders, discounts for flexible delivery

AI Model:

- Demand forecasting (warehouse stocking)
- Route optimization (traveling salesman problem)
- Dynamic pricing algorithm

Success Metric: Delivery cost reduced 2.5% → 2%, +20% orders/driver

Month 9: Intelligent Notifications (Zero Spam)

What: AI sends notifications only when helpful (not annoying) **Why:** Most apps over-notify → users ignore → churn

Features:

-  **Smart timing:** Notify when shop owner is most active (ML-based)
-  **Personalized:** "Your top 3 products are low, reorder now?"
-  **Actionable:** Every notification = 1-tap action

-  **Frequency limit:** Max 2 notifications/day (unless urgent)
-  **Opt-out friendly:** Easy to customize notification preferences

AI Model:

- User behavior prediction (when to send notifications)
- NLP for message personalization
- A/B testing for message optimization

Success Metric: 60% notification click-through rate (vs 5-10% industry avg)

Phase 3 Goal: 60% of orders fully automated (AI end-to-end)

PHASE 4: ADVANCED AI (Months 10-12)

Predictive & Prescriptive

Month 10: Price Intelligence

What: AI alerts shop owners when competitor prices change **Why:** Help shops stay competitive

Features:

-  AI monitors competitor pricing (nearby shops, online)
-  Alerts: "Your rice is 5% more expensive than nearby shop"
-  Suggests: "Lower price or highlight quality difference"
-  Dynamic discounts: AI recommends promotions to win back customers

Success Metric: Shops using price intelligence have 10% higher retention

Month 11: Business Insights Dashboard

What: AI analyzes shop performance → Actionable insights **Why:** Turn data into business growth

Features:

-  "Your snack sales are up 30% - stock more"
-  "You're losing customers on Mondays - why?"

-  "Shops like yours sell [Product X] - try it?"
-  Revenue forecasting: "At this rate, you'll earn X this month"
-  Competitor benchmarking: "You're in top 20% performers"

Success Metric: Shops using insights grow revenue 25% faster

Month 12: Multi-Shop Management (For Growing Shop Owners)

What: Shop owners with 2-5 shops manage all from one app **Why:** Many successful owners expand to multiple locations

Features:

-  Unified dashboard (view all shops)
-  AI allocates inventory across shops
-  Consolidated ordering (bulk discounts)
-  Performance comparison (which shop performs best?)
-  Staff management (assign roles per shop)

Success Metric: 10% of customers manage 2+ shops via app

Phase 4 Goal: Aladin AI = Indispensable business tool (not just ordering app)

TECHNICAL ARCHITECTURE

Frontend (Mobile App)

Technology: React Native (iOS + Android)

Why: Cross-platform, fast development, Zalo integration

Key Screens:

- └ Home: Quick reorder (1-tap popular items)
- └ Voice Order: Mic button (speak to order)
- └ Photo Order: Camera (snap invoice → order)
- └ Order History: Track deliveries
- └ Recommendations: AI-suggested products
- └ Insights: Business analytics dashboard

Backend (API + AI Services)

API: Node.js + Express (fast, scalable)

Database: PostgreSQL (relational data) + Redis (caching)

AI Services:

- └ Speech-to-Text: Zalo API + Google Cloud
- └ OCR: PaddleOCR (Vietnamese) + Google Vision
- └ NLP: spaCy Vietnamese + Claude API (LLM)
- └ Forecasting: Python (Prophet, scikit-learn)
- └ Recommendations: Python (collaborative filtering)
- └ Route Optimization: OR-Tools (Google)

Hosting: AWS or Google Cloud

CDN: Cloudflare (fast delivery)

AI/ML Infrastructure

Training: Python + Jupyter + TensorFlow/PyTorch

Deployment: FastAPI (ML model serving)

Monitoring: MLflow (track model performance)

Data Pipeline: Airflow (daily training jobs)

Feature Store: Feast (ML features)

Integrations

Zalo: Mini App + Official Account + ZaloPay

Payment: ZaloPay + Momo + Bank Transfer

Logistics: In-house + Grab (overflow)

ERP: Custom inventory management

CRM: Customer database (PostgreSQL)

AI MODELS ROADMAP

Phase 1 (Months 1-3): Rules-Based

- No ML yet, use if-then logic
- Build data pipeline (collect order history, user behavior)
- **Goal:** Gather 10,000+ orders for training data

Phase 2 (Months 4-6): Basic ML

Model 1: Demand Forecasting

- Algorithm: Prophet (time-series)
- Input: Order history (3+ months)
- Output: Predicted stockout date
- Training: Weekly retraining

Model 2: Product Recommendations

- Algorithm: Collaborative filtering (matrix factorization)
- Input: Order patterns across customers
- Output: Top 5 recommended products
- Training: Daily retraining

Model 3: Intent Classification (Voice/Chat)

- Algorithm: spaCy Vietnamese + custom NER
- Input: User message (text/voice)
- Output: Intent (order, track, ask question) + Entities (product, quantity)
- Training: Fine-tune on 1,000+ labeled conversations

Phase 3 (Months 7-9): Advanced ML

Model 4: Auto-Reorder Optimizer

- Algorithm: Reinforcement learning (Q-learning)
- Input: Inventory level, order history, delivery time
- Output: Optimal reorder timing + quantity
- Training: Simulated environment, then A/B test

Model 5: Route Optimization

- Algorithm: OR-Tools (vehicle routing problem)
- Input: Delivery addresses, time windows, vehicle capacity
- Output: Optimized routes
- Training: Daily optimization

Model 6: Dynamic Pricing

- Algorithm: Multi-armed bandit (A/B testing)
 - Input: Demand, competitor prices, customer segments
 - Output: Optimal price per product per customer
 - Training: Real-time learning
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Phase 4 (Months 10-12): LLM + Predictive

Model 7: Conversational AI (LLM)

- Algorithm: Claude API (fine-tuned on Vietnamese + Aladin data)
- Input: Multi-turn conversation
- Output: Natural language response + action (place order, suggest product)
- Training: Fine-tune on 10,000+ conversations

Model 8: Business Insights (Prescriptive Analytics)

- Algorithm: Causal inference + XGBoost
 - Input: Shop performance data, external factors (weather, holidays)
 - Output: Actionable recommendations
 - Training: Monthly retraining
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DATA STRATEGY

What Data to Collect (Privacy-First)

Customer Data:

- Shop profile (location, type, size)
- Order history (products, quantities, frequency)
- Payment methods (preferences)
- App usage (features used, time spent)

Product Data:

- Catalog (SKUs, categories, brands)
- Pricing (historical, competitor)

- Inventory levels (warehouse + customer)
- Demand patterns (seasonality, trends)

Operational Data:

- Delivery routes (times, distances)
- Driver performance (speed, accuracy)
- Warehouse operations (stock in/out)
- Customer support (issues, feedback)

AI Training Data:

- Voice recordings (with consent, anonymized)
- OCR images (invoices, orders)
- Conversation logs (intent, entities)
- Model predictions (accuracy tracking)

Privacy Compliance:

-  Get user consent for voice/image data
 -  Anonymize sensitive data (names, addresses)
 -  Secure storage (encryption at rest)
 -  Allow users to delete their data (GDPR-like)
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DEVELOPMENT TIMELINE

Sprint Structure (2-Week Sprints)

Months 1-3: 6 sprints

- Sprint 1-2: Zalo chatbot MVP
- Sprint 3-4: Voice input (speech-to-text)
- Sprint 5-6: OCR (photo-to-order)

Months 4-6: 6 sprints

- Sprint 7-8: Demand forecasting
- Sprint 9-10: Recommendations engine
- Sprint 11-12: Conversational AI (voice agent)

Months 7-9: 6 sprints

- Sprint 13-14: Auto-reorder system
- Sprint 15-16: Fulfillment optimization
- Sprint 17-18: Smart notifications

Months 10-12: 6 sprints

- Sprint 19-20: Price intelligence
 - Sprint 21-22: Business insights dashboard
 - Sprint 23-24: Multi-shop management
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TEAM REQUIREMENTS

Current Team (Month 1-3)

- **Karthick (CTO):** Architecture, backend, AI strategy
- **1 Mobile Developer:** React Native (Zalo integration)
- **1 Backend Developer:** Node.js API
- **1 AI/ML Engineer (Part-time/Contractor):** OCR, NLP

Total: 3-4 people

Scaled Team (Month 4-9)

- **Karthick (CTO):** Strategy, architecture oversight
- **2 Mobile Developers:** iOS + Android features
- **2 Backend Engineers:** API scaling, database optimization
- **2 AI/ML Engineers:** Forecasting, recommendations, LLM
- **1 Data Engineer:** Data pipeline, feature engineering
- **1 QA Engineer:** Testing, bug fixing
- **1 Product Manager:** Roadmap, user feedback

Total: 9-10 people

Full Team (Month 10-12)

- Add: 1 DevOps Engineer (infrastructure scaling)
- Add: 1 UI/UX Designer (polish, user research)

Total: 11-12 people

SUCCESS METRICS (By Phase)

Phase 1 (Months 1-3)

-  60% of orders via Zalo (vs phone)
 -  20% of orders via voice
 -  15% of orders via photo OCR
 -  85% customer satisfaction with app
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Phase 2 (Months 4-6)

-  70% demand forecast accuracy
 -  30% customers adopt auto-suggestions
 -  AOV increase: +15% (recommendations)
 -  50% of orders via voice AI (conversational)
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Phase 3 (Months 7-9)

-  40% of shops on auto-reorder
 -  Delivery cost reduced: 2.5% → 2%
 -  60% notification click-through rate
 -  60% of orders fully automated
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Phase 4 (Months 10-12)

-  80% of customers use business insights
-  Shops using AI grow 25% faster
-  10% of customers manage 2+ shops
-  Aladin AI = #1 rated B2B commerce app in Vietnam

INVESTMENT REQUIRED

Phase 1 (Months 1-3): \$30-40K

- Team: \$25K (3-4 people, part contractors)
- Infrastructure: \$3K (cloud hosting, APIs)
- Third-party: \$2K (Zalo, Google OCR)

Phase 2 (Months 4-6): \$60-80K

- Team: \$50K (6-7 people)
- Infrastructure: \$5K (scaling, ML servers)
- Third-party: \$5K (Claude API, data labeling)

Phase 3 (Months 7-9): \$80-100K

- Team: \$70K (9-10 people)
- Infrastructure: \$8K (auto-scaling, ML ops)
- Third-party: \$5K (LLM API, advanced features)

Phase 4 (Months 10-12): \$100-120K

- Team: \$85K (11-12 people)
- Infrastructure: \$10K (high availability)
- Third-party: \$5K (analytics, integrations)

Total 12-Month Investment: \$270-340K

COMPETITIVE ADVANTAGE

Why Aladin AI Will Win

1. Vietnamese-First Design

- Competitors use English-first AI (poor Vietnamese support)
- We optimize for Vietnamese dialects, slang, OCR

2. Rural-Optimized

- Low-bandwidth app (works on 3G)
- Offline mode (order queue syncs later)
- Simple UI (designed for 50+ year-old shop owners)

3. Voice-First (Not Text-First)

- Rural shop owners prefer speaking over typing
- 10 seconds to order vs 5 minutes typing

4. Context-Aware AI

- Learns individual shop preferences
- Understands "order the usual" without clarification

5. Full-Stack Automation

- Not just ordering - entire supply chain AI-optimized
- Competitors focus on catalog, we focus on workflow

RISKS & MITIGATION

Risk 1: AI Accuracy Too Low (Users Frustrated)

Mitigation:

- Phase 1: Human-in-the-loop (fallback to manual)
- Phase 2: 85% accuracy threshold before full rollout
- Always offer "correct this" option

Risk 2: Users Don't Trust AI (Prefer Humans)

Mitigation:

- Gradual rollout: Suggest, don't force
 - Show AI as "assistant" not "replacement"
 - Transparency: "AI suggested this because..."
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Risk 3: Technical Complexity (Delays)

Mitigation:

- Start simple (rules-based), add AI incrementally
 - Use pre-trained models (Claude, Google) before building custom
 - Hire experienced AI engineers early
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Risk 4: Cost of AI (LLM APIs Expensive)

Mitigation:

- Cache responses (reduce API calls)
 - Use cheaper models for simple tasks (spaCy)
 - Self-host models when volume justifies (Month 9+)
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THE VISION (3 Years)

2026: Aladin AI = Best B2B commerce app in Vietnam

- 10,000 shops using AI ordering
- 70% of orders fully automated
- \$1M+ monthly GMV via app

2027: Expand to Indonesia, Philippines

- Clone Vietnamese playbook
- Localize AI for Bahasa, Tagalog
- 100,000 shops across Southeast Asia

2028: AI Platform for All Traditional Trade

- Not just FMCG - expand to electronics, fashion, hardware

- Become "Shopify + Pinduoduo + AI" for emerging markets
 - 1M+ shops, \$100M+ monthly GMV
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IMMEDIATE NEXT STEPS (This Month)

Week 1: Proof of Concept

- ☐ Karthick: Set up Zalo Mini App sandbox
- ☐ Test: Basic chatbot (menu-based ordering)
- ☐ Test: Zalo voice message → text conversion

Week 2: Vietnamese OCR Test

- ☐ Collect: 50 real invoices from customers
- ☐ Test: Google Vision + PaddleOCR accuracy
- ☐ Build: MVP product matching algorithm

Week 3: AI Strategy Workshop

- ☐ Team meeting: Prioritize features (voice vs OCR first?)
- ☐ Decision: Build vs buy (Claude API vs custom LLM?)
- ☐ Hire: Find AI/ML engineer (contractor to start)

Week 4: Launch Plan

- ☐ Recruit: 20 beta testers (high-trust customers)
 - ☐ Timeline: Month 1-3 roadmap locked
 - ☐ Budget: Secure \$30-40K for Phase 1
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READY TO BUILD THE FUTURE OF RURAL COMMERCE? 🚀